

README

CRM/ERP Platform Documentation

CRM/ERP Platform — Documentation

A metadata-driven, multi-tenant enterprise CRM/ERP system built with **Next.js 16** (frontend) and **NestJS** (backend).

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Configuration	Environment variables reference
Deployment	Production deployment steps
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Architecture & Design

Document	Description
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Project-Structure	Directory layout, file conventions
Database	Schema, models, migrations
Authentication	JWT auth, portal auth, sessions
Permissions	Role-based and field-level access control

Document	Description
Security	Security model, CORS, rate limiting

Features & Modules

Document	Description
Modules	Dynamic module (table) system
Form-Builder	Public forms, OCR, submission receipts
Workflow	Automation engine, triggers, actions
Reporting	Analytics, dashboards, saved reports
Pivot-Table	Cross-tab analysis
OCR	Document extraction via Anthropic Claude
File-Storage	File upload, serving, gallery

Operations

Document	Description
API	Full REST API reference
Backup-Recovery	Backup strategies, restore procedures
Troubleshooting	Common issues and fixes
FAQ	Frequently asked questions

Project

Document	Description
Changelog	Version history
Future-Roadmap	Planned features

Quick Start

```
# 1. Clone and install
git clone <repo> crm && cd crm
cd backend && npm install
cd ../mycrm && npm install

# 2. Configure
cp backend/.env.example backend/.env    # edit DATABASE_URL, JWT_SECRET, etc.
cp mycrm/.env.local.example mycrm/.env.local

# 3. Database
cd backend
npx prisma migrate deploy
npx prisma generate

# 4. Run (two terminals)
cd backend && npm run start:dev
cd mycrm    && npm run dev
```

Open <http://localhost:3000> — create your organization on first launch.

Tech Stack

Layer	Technology
Frontend	Next.js 16 (App Router), React 19, TypeScript
Backend	NestJS 10, TypeScript
Database	MySQL 8 via Prisma ORM
Auth	JWT (24h access / 7d refresh), bcryptjs
State	Zustand (13 stores)
Drag-drop	@dnd-kit/core, @dnd-kit/sortable
Charts	Recharts

Layer	Technology
Tables	@tanstack/react-table
Real-time	Socket.io (WebSocket)
Queue	BullMQ + Redis
AI/OCR	Anthropic Claude API (claude-sonnet-4-6)
Email	Nodemailer (custom SMTP)
Calendar	Google Calendar OAuth2
Storage	Local disk (backend/uploads/)
Deployment	Railway (backend), Netlify (frontend)